

ELECTRICAL AND CURRENT TRANSPORT PROPERTIES OF Au/CoPc/UNDOPED-InP METAL/POLYMER/SEMICONDUCTOR (MPS) STRUCTURE

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ABSTRACT

In the present work, the electrical and current transport properties of Au/undoped-InP Schottky diode (SD) and Au/CoPc/undoped-InP metal/polymer/semiconductor (MPS) structure are investigated using current-voltage (I-V) method at room temperature. The MPS structure exhibited low reverse leakage current and high barrier height compared to the Schottky diode (SD). The barrier height (BH) and ideality factor (n) values of SD and MPS are found to be 0.66 eV, 1.58 and 0.77 eV, 1.42, respectively. The Norde method is used to determine the series resistance (R_s) and BH values, and the corresponding values are 0.64 eV and 22.4 k Ω , 0.78 eV and 1.8 M Ω for the SD and MPS structure, respectively. The surface potential method was also used to determine the BH and ideality factor values and the values are 0.64 eV, 2.84 for the SD and 0.77 eV and 1.98 for the MPS structure, respectively. The BH is extracted using the I-V, Cheung's function, Norde method and ψ_s -V plot, and the BH and ideality factor values are comparable with one each other, which indicates their steadiness and validity. The determined interface state density (N_{ss}) for the MPS structure is less than the SD, suggesting that the CoPc layer has effectively passivated the surface of the undoped-InP, thus the N_{ss} value. The reverse leakage current mechanism of SD and MPS structure are analysed using Poole-Frenkel and Schottky emission. Results demonstrate that the Schottky emission dominates the reverse leakage current in the SD and MPS structure. The results suggest that the CoPc material can be used in the fabrication of MPS devices.

Keywords: Cobalt Phthalocyanine, Density of states, undoped-InP, Electrical and Current transport properties.
